# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The radius of a sphere is multiplied by $6$. Its surface area becomes ___ times: 1

**(a)** 216 **(b)** 6 **(c)** 36 **(d)** 12

**2.** The edge of a cube is multiplied by $5$. Its volume becomes ___ times: 1

**(a)** 25 **(b)** 125 **(c)** 15 **(d)** 5

**3.** The radii of two spheres are in the ratio $1:6$. The ratio of their volumes is: 1

**(a)** $1:36$ **(b)** $1:216$ **(c)** $1:210$ **(d)** $1:6$

**4.** Take $\pi = \frac{22}{7}$. The total surface area of a cone of radius $10.5$ cm and slant height $17.5$ cm is: 1

**(a)** $1017$ cm$^2$ **(b)** $924$ cm$^2$ **(c)** $1110$ cm$^2$ **(d)** $831$ cm$^2$

**5.** A cone has radius $8$ cm and height $15$ cm. Its slant height is: 1

**(a)** $19$ cm **(b)** $18$ cm **(c)** $16$ cm **(d)** $17$ cm

**6.** A tank measures $9$ m $\times$ $7$ m $\times$ $11$ m. Its capacity in litres ($1\ \text{m}^3 = 1000$ L) is: 1

**(a)** $6930000$ **(b)** $693$ **(c)** $69300$ **(d)** $693000$

**7.** The number of cubes of edge $5$ cm that can be cut from a cuboid $30$ cm $\times$ $15$ cm $\times$ $20$ cm is: 1

**(a)** $71$ **(b)** $73$ **(c)** $78$ **(d)** $72$

**8.** A pyramid has a square base of side $12$ cm and height $24$ cm. Its volume is: 1

**(a)** $1268$ cm$^3$ **(b)** $1152$ cm$^3$ **(c)** $1384$ cm$^3$ **(d)** $1036$ cm$^3$

**9.** The radius of a sphere is multiplied by $4$. Its surface area becomes ___ times: 1

**(a)** 64 **(b)** 8 **(c)** 4 **(d)** 16

**10.** The edge of a cube is multiplied by $6$. Its volume becomes ___ times: 1

**(a)** 36 **(b)** 216 **(c)** 6 **(d)** 18

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Take $\pi = \frac{22}{7}$. A cylinder of height 6 cm has volume 2079 cm$^3$. Find its radius and its curved surface area. 2

**12.** The total surface area of a cube is 294 cm$^2$. Find its edge, its volume and the length of its face diagonal in terms of $\sqrt{2}$. 2

**13.** How many cubes of edge 8 cm can be cut from a cuboid measuring 24 cm × 40 cm × 16 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 2

**14.** Take $\pi = \frac{22}{7}$. A cylinder of height 24 cm has volume 14784 $cm^3$. Find its radius and its curved surface area. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Take $\pi = \frac{22}{7}$. A cone has radius 10.5 cm and height 14 cm. Find its slant height, curved surface area and volume. 3

**16.** The edge of a cube is 9 cm. If every edge is multiplied by 3, find the new surface area and volume and the factors by which they change. 3

**17.** Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 21 cm. 3

OR

Take $\pi = \frac{22}{7}$. A cone has radius 28 cm and height 52.5 cm. Find its slant height, curved surface area and volume. 3

**18.** The edge of a cube is 4 cm. If every edge is multiplied by 2, find the new surface area and volume and the factors by which they change. 3

OR

Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area of a solid hemisphere of radius 35 cm. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A cylindrical metal block of radius 7 cm and height 24 cm is melted and recast into cones of the same radius and height 1 cm. How many cones are made? 5

OR

A solid cuboid 50 cm × 35 cm × 40 cm is melted and recast into cubes of edge 5 cm. Find the number of cubes and the total surface area of all the cubes. 5

**20.** A cylindrical metal block of radius 14 cm and height 24 cm is melted and recast into cones of the same radius and height 1 cm. How many cones are made? 5

OR

A solid cuboid 50 cm × 15 cm × 15 cm is melted and recast into cubes of edge 5 cm. Find the number of cubes and the total surface area of all the cubes. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**

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